

National University of Computer and Emerging Sciences, Lahore Campus

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	Paper Date:	Due Before Mid-II exam	Weight	3%
	Section:	A,B	Page(s):	
	Exam Type:	Mid-II Review Assignment		

Q No 1 [Perceptron and Perceptron Learning]

Figure 1 show a single perceptron with two inputs x_1 and x_2 and a bias term that is permanently set to.

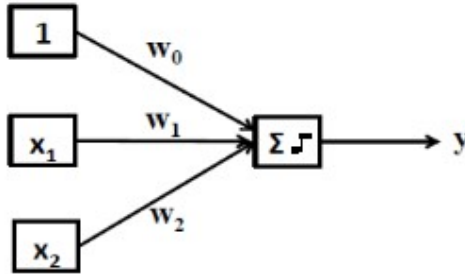


Figure 1

Output of this perceptron is computed using a simple threshold and is given as

$$y = \begin{cases} 1 & \text{if } (w_0 + w_1 \cdot x_1 + w_2 \cdot x_2) \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

- It is desirable to learn weights $[w_0 \ w_1 \ w_2]$ of a perceptron that is similar to the perceptron in Figure 1. This perceptron will have two real valued inputs x_1 and x_2 and a bias term set permanently to 1. The objective is to learn weights such that perceptron gives a 1 as output if $x_2 \geq x_1$ and -1 otherwise. In this part you are required to create training data that can be used to learn weights of the perceptron. The training data must consist of two positive examples (i.e. examples for which output must be 1) and two negative examples (i.e. examples for which output must be -1)
- Use perceptron learning rule to learn weights $[w_0 \ w_1 \ w_2]$ of the perceptron using your training data of part a. You must start with initial weights $[0 \ 0 \ 0]$ and only do one iteration/epochs for learning the weights

Q No 2 [Search Solution Using GA]

Subset-sum is an important NP-complete decision problem in computer science and can be stated as follows

For a multi-set of integers, is there a non-empty subset whose sum is exactly equal to a given value S .

For example, given the multi-set $\{-7, -7, -3, -2, 9000, 5, 8\}$ of seven integers and the value -19, the answer to the subset sum problem is **YES** because the multi-subset $\{-7, -7, -3, -2\}$ sums to -19.

Similarly for the given set $\{-7, -3, -2, 9000, 5, 8\}$ of six integers and the value -5, the answer to the subset sum problem is **YES** because several subset exists that sums to -5 like $\{-3, -2\}$ and $\{-7, -3, 5\}$ etc.

Furthermore for the same set {-7, -3, -2, 9000, 5, 8} of six integers and the value 18000, the answer to the subset sum problem is **NO** because no subset exists that sums to 18000.

For a set having n-elements we can represent the subset sum problem as an evolutionary search problem using the following encoding of solution/chromosome

Chromosome: is a bit vector of length n with a bit being set to 1 if the element is part of the subset and 0 otherwise.

For example, for the multi-set {-7, -7, -3, -2, 9000, 5, 8} the chromosome will be a bit vector of length 7, hence the chromosome 1111000 will represent the solution/subset {-7, -7, -3, -2}. Similarly the chromosome 1111111 represents the solution/subset {-7, -7, -3, -2, 9000, 5, 8}

Chromosome Fitness: The fitness of a solution can then be computed using $1 / (|S - \sum| + 1)$ where S is the required value of sum and $\sum$ is sum of the subset and $|x|$ represents absolute value of x.

For example, for the multi-set {-7, -7, -3, -2, 9000, 5, 8} and $S = -19$ the chromosome 1111000 has fitness value of **1** whereas the chromosome 1111111 has fitness 1/8995. Verify that you understand the formula for computing fitness.

Assume that in an implementation of GA is being used to find a subset of the set {2, 3, 4, 8, 16} having sum **17** and that the following five solutions have been randomly generated as the initial population.

{11001, 10001, 00111, 011111, 00010}

Part a) Compute fitness of each solution in this population using the formula given above.

Part b) Use fitness proportionate selection to select 1 elite/solution that must be transferred unchanged to the next generation of solutions.

Part c) Generate the remaining chromosomes/solutions of the next generation using single point crossover and mutation operation. Here assume that GA uses fitness proportionate selection and that the mutation rate is 0.001. Points will be awarded for correct selection (fitness-proportionate), correct crossover and correct mutation and correct computation of next generation

Whenever you need a random number select the next random number from the following **list/2D-array** of randomly generated numbers and if the list exhausts and you still need more random numbers then you must start choosing the number from the beginning (**row 0, col 0**) of the list again.

To make the things a bit more interesting and more specific to each students, you are required to **start choosing the random numbers from different place in the list** based on the last digit of your registration number.

So if the last digit of your registration number (XXX-XXX0) is 0 you must start at position (0, 0), shown in bold and green color, and move on row-wise until all rows have been used.

Similarly if the last digit of your registration number (XXX-XXX5) is 5 then you must begin at position (0, 5), shown in bold and red color, and move on till the end.

R\C	0	1	2	3	4	5	6	7	8	9
0	0.487	0.937	0.291	0.816	0.051	0.621	0.248	0.691	0.882	0.025
1	0.038	0.096	0.322	0.179	0.967	0.064	0.930	0.014	0.091	0.844
2	0.766	0.358	0.797	0.609	0.846	0.498	0.612	0.476	0.946	0.873
3	0.757	0.944	0.749	0.966	0.033	0.755	0.659	0.161	0.684	0.095

Q No 2 [Neural Networks]

Consider a simple feedforward neural network (MLP) with two inputs, 1 hidden layer with 2 neurons and ReLU activation, and an output layer with a single neuron with sigmoid activation. The weights and biases of the trained network as follows:

Hidden Layer Weights:		Hidden layer Biases		Output Layer Weights:		output layer Bias
0.5	-1	0	1	1	-1	0.5
1.5	0	1				

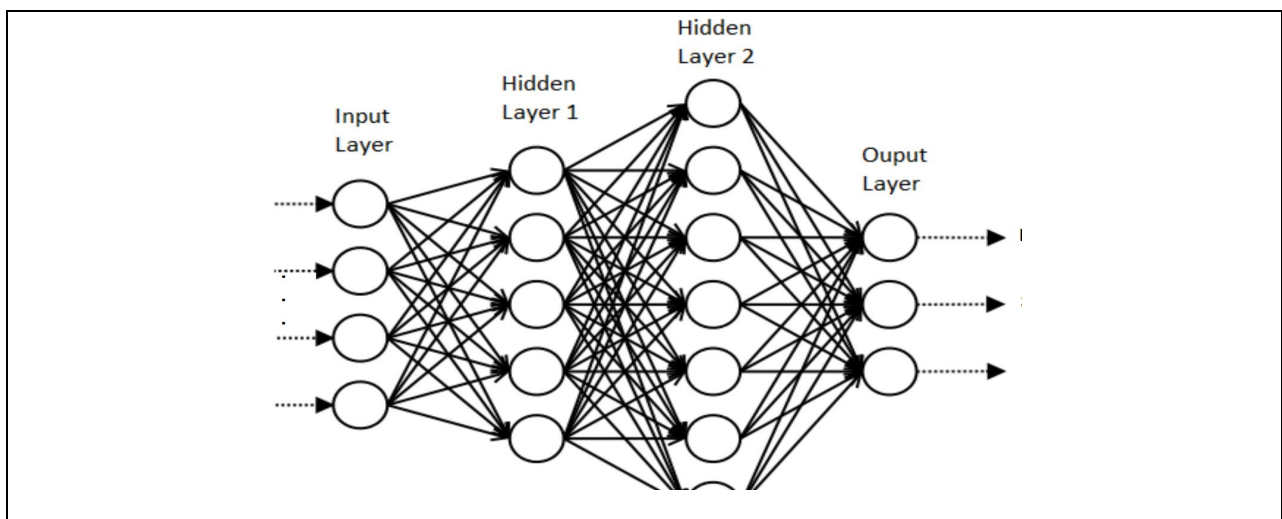
Perform the **forward pass** and compute the final **output** of the network if input is $[2 \ 1]^T$

Q No 3 [Neural Networks]

Inspired by the movie WALL-E a teacher at FAST-LHR is designing a line following robot. Being aware of the representation and learning power of a feed-forward artificial neural network, he decided to use such a neural network with two hidden layers of neurons for creating the main controller/brain of the robot. The initial prototype of the robot will only follow smooth lines i.e. lines without any sharp turns.

Input to the controller/brain of this robot will be a grey-scale camera image of dimensions $N \times N$. The number of neurons in the first hidden layer will be H_1 and the number of neurons in the second hidden layer will be H_2 whereas there will be **3** neurons in the output layer with each output being **between 0 and 1** specifying the direction in which the robot need to turn. The robot will be turned slightly towards left, right or will go straight depending upon the largest value of the output neuron **Left, Straight or Right** respectively.

Such a neural network is depicted in the figure below



In this question we will assume that **N** is equal to the last four digit number in you university ID whereas **H₁** is the **number consisting of the year/batch number** in your ID and **H₂** is the **most significant two digits of N**. For example if your id is L17-4526 then N is 4526, H₁ is 17 and H₂ is 45.

Part a) [Neural Network Weights]

Specify the number of weights needed to create such a neural network. Assume that each neuron in this network has a bias term as well. Further if we assume that each weight is a double precision number represented using 8-bytes, approximately how many kilo bytes will be needed to store this neural network on some storage.

Part b) [Neural Network Computations]

Assuming that the camera is generating 30 frames/Images per second and the neural network will be used to make a decision for each one of the frames, how many floating point multiplications and additions will be computed per second if this network is used for making decision.

Part c) [Neural Network Coding]

Write a C++/Python function that can be used to compute output of a single network layer. Assume that the activation function is **sigmoid** for each neuron of the layer and that the weights of neurons of that layer are stored in a global array of weights **W** having dimensions **H x N** i.e. **H** neurons each has **N** weights. Your function will have only one parameter i.e. a vector/array **X** of inputs and will return a vector/array **O** of computed outputs.

Part d) [Preparing Training Data]

We know that weights of a neural network are typically learned using back-propagation learning algorithm which is a supervised learning algorithm.

- i) What do we mean by supervised learning? Give names of three supervised learning algorithms for classification.
- ii) Briefly describe some method to prepare training data for learning weights of the network described in the previous parts.

Evaluation Metrics for Classification and Regression

4. Define the purpose of training data, validation data and test data used for training a classification model.

Use online sources to read about the commonly used evaluation measures for classification including Accuracy, Precision, Recall and F1 measure and answer the following questions.

5. Provide an example of a binary classification problem where

- a. False negatives are more costly than false positives
 - b. False positives are more costly than false positives
6. For a binary classification problem which metric is most appropriate when false positives are more costly than false negatives?
7. A binary classifier was tested on a dataset of 100 samples. The results are summarized in the following confusion matrix:

	Predicted: Positive	Predicted: Negative
Actual: Positive	36	9
Actual: Negative	15	40

- a. Using the confusion matrix above, calculate True Positives (TP) , False Positives (FP), True Negatives (TN), False Negatives (FN), Precision, Recall, Accuracy, and F1-score

Use online sources to read about the commonly used evaluation measures for regression including MSE, RMSE and R²-score and answer the following questions.

8. Which regression metric is most **sensitive to outliers**?
- a. Mean Absolute Error (MAE)
 - b. Root Mean Squared Error (RMSE)
 - c. R² Score
9. Given the actual and predicted values below:

Actual	Predicted
10	12
15	14
20	25

Calculate the MSE and MAE for the predictions.